

Assignment 2: Numerical Integration

1. Implement the multiple segment version of the trapezoidal and Simpson's 1/3 rules for a *line integral* with limits from a to b as a MATLAB function. Inputs should be integration limits (a, b), number of segments (n) (check for evenness of n for Simpson's 1/3 rule), and a function name (f). Test this implementation for both methods by computing the following line integral by using $n = 10, 100$, and 1000 :

$$\int_0^1 x^{0.1} (1.2 - x) (1 - e^{20(x-1)}) dx$$

Also use the MATLAB's **integral** function to evaluate the same integral. Evaluate the percent error in each case based on the true solution of 0.602297. Note that you can use the MATLAB help utility for assistance regarding usage of these functions and other functions.

2. An 12-m beam is subjected to a load, and the shear force follows the following equation

$$V(x) = 5 + 0.25x^2$$

Where V is the shear force and x is the length in distance along the beam. We know that $V = dM / dx$ and M is the bending moment. Integration yields the following relationship

$$M = M_0 + \int_0^x V dx$$

If M_0 is zero and $x = 12$ m, calculate M using (a) analytical integration, (b) trapezoidal rule, and (c) Simpson's rule. For (b) and (c) use 0.5-m increments.

3. The force on a sailboat mast can be represented by the following function:

$$f(z) = 200 \left(\frac{z}{7+z} \right) e^{-2.7z/H}$$

Where z = the elevation above the deck and H = the height of the mast. The total force F exerted on the mast can be determined by integrating this function over the height of the mast:

$$F = \int_0^H f(z) dz$$

The line of action at height d above the deck can also be determined by integration:

$$d = \frac{\int_0^H z f(z) dz}{\int_0^H f(z) dz}$$

Use Trapezoidal and 1/3 Simpson's rules to compute F and d for the case $H = 30$. Use 100 segments for both cases ($n = 100$). Compare your solutions with those from MATLAB's **integral** function.

4. Velocity distribution of a fluid undergoing turbulent flow in a pipe is given by

$$v = 2 \left(1 - \frac{r}{R} \right)^{1/6}$$

Where R is the radius of the pipe ($= 2$ cm) and r is the radius at which the velocity is computed. Determine the flow rate Q through the pipe using Simpson's 1/3 rule and compare it with the solution from MATLAB's '**integral**' function and analytical solution using the symbolic '**int**' function. Flow rate Q is given by

$$Q = 2\pi \int_0^R r v dr$$

5. The following data was collected for a cross-section of a river (y = distance from the bank, H = depth and U = velocity):

y, m	0	1	3	5	7	8	9	10
H, m	0	1	1.5	3	3.5	3.2	2	0
U, m	0	0.1	0.12	0.2	0.25	0.3	0.15	0

Use numerical integration to compute the (a) cross-sectional area (A_c), (b) average depth, (c) flow rate, and (d) the average velocity. Note that the cross-sectional area (A_c) and the flow rate (Q) can be computed as

$$A_c = \int_0^b H(y) dy, \quad \bar{H} = \frac{A_c}{b}, \quad Q = \int_0^b H(y) U(y) dy, \quad \bar{U} = \frac{Q}{A_c}$$

Where b ($=10$ m) is the width of the river.

6. Extend your code in Q.1 to perform double integrals and then evaluate the following double integral using trapezoidal rule and Simpson's 1/3 rule with $n = 100$.

$$\int_{x=-2}^{x=2} \int_{y=0}^{y=4} (x^2 - 3y^2 + xy^3) dy dx$$

Compare your answers with the MATLAB function '**integral2**' and analytical solution using the symbolic '**int**' function.

7. Write a MATLAB code to perform general higher dimensional integrals using the recursive functionality. Then evaluate the following triple integral using Trapezoidal rule and Simpson's 1/3 rule with $n = 20$.

$$\int_{x=-4}^{x=4} \int_{y=0}^{y=6} \int_{z=-1}^{z=3} z(x^3 - 2xy - z) dz dy dx$$

Compare your answer with the MATLAB function '**integral3**' and analytical solution using the symbolic '**int**' function.

8. Develop a function to implement a Monte-Carlo integration of a general multiple integral containing m integrals. The formula for Monte-Carlo integration for a 3D problem ($m=3$) is given by

$$I \approx \frac{\mu(D)}{n} \sum_{k=1}^n f(x_k, y_k, z_k)$$

where $\mu(D)$ is the volume of the domain D . Assuming a Cartesian coordinate system $\mu(D)$ is given by

$$\mu(D) = \int_a^b \int_c^d \int_e^f dz dy dx = (b-a)(d-c)(f-e)$$

In the above expression we assume that the shape of the domain is regular; i.e., a cuboid. This is a reasonable assumption to use. For irregular domains we can use a regular domain that encloses the irregular domain; function value is set to zero outside the irregular domain. Compute the triple integral in [Question 7](#) using Montecarlo integration using $n = 1000$ points and compute the error. What happens to the error if you increase n to 100,000? *Hint:* $\mu(D)$ can be computed using the ‘**prod**’ function in MATLAB. Use the ‘**rand**’ function to generate the uniformly distributed random numbers.